

WordNet analysis

WordNet is a manually constructed semantic network whose main building blocks are sets of words and phrases that were considered “cognitively synonymous” by the WordNet compilers, so-called synsets (Fellbaum 2006). For example, *sleep* is associated with a number of synsets, shown in (1a). The synsets are connected by hyperonym and hyponym relations.

- (1) a. **Synsets associated with *sleep***
 sleep.n.01 {sleep,slumber}
 sleep.n.02 {sleep,sopor}
 sleep.n.03 {sleep,nap}
 rest.n.05 {rest,eternal_rest,sleep,eternal_sleep,quietus}
 sleep.v.01 {sleep,kip,slumber,log_Z's,catch_some_Z's}
 sleep.v.02{sleep}
- b. **Synsets associated with *snooze***
 nap.n.04 {nap,catnap,cat_sleep,forty_winks,short_sleep,snooze}
 snooze.v.01 {snooze,drowse,doze}
- c. **Synsets associated with *snore***
 snore.n.01{snore}
 snore.n.02{snore,snoring}
 snore.n.02{stertor}
 snore.v.01{snore,saw_wood,saw_logs}

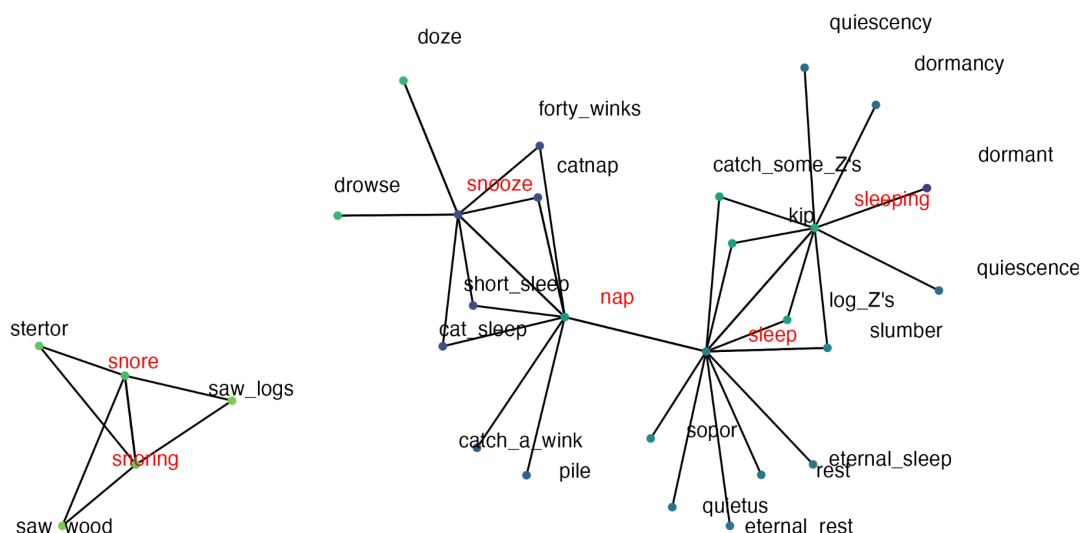


Fig. X: Example for connections between lemmas in WordNet. The node colors show the synset to which the lemmas belong. The red colors indicate the target words for which the corresponding synsets were extracted.

We used two pertinent measures to determine the similarity between the synsets associated with the participants' responses and the target word¹: firstly, path similarity, which is calculated by determining the shortest path between two synsets (p) as shown in (2a); secondly, the Wu-Palmer similarity, which makes use of the lowest lowest common hyponym of two synsets (the so-called least common subsumer or LCS). It is calculated as in (2b), where i is the shortest path distance from the first synset to the LCS, j is the shortest path distance from the second synset to the LCS, and k is the number of nodes from the LCS to the root node, i.e. an abstract superordinate concept.²

- (2) a. Path similarity: $\frac{1}{p+1}$
- b. Wu-Palmer similarity: $\frac{2k}{i+j+2k}$

Words for which no synset is available in WordNet were not taken into account in this analysis (e.g. onomatopoeia like *uuuuuuuuahhhhh* or *plumplumploomplommploom* which sometimes were given as answers by the participants). Fig. 1 shows the results. The words printed in blue are the participants' answers (see the supplementary material for a zoomable version of the figure), the boxplots show the distribution of the path distance and the WUP distance, respectively. As the path similarity is based on the shortest path between two synsets, while Wu-Palmer is based on a common hyponym, it is not surprising that the values obtained with the former are much lower than the Wu-Palmer distance values. While our manual annotation suggests that the participants' answers are semantically very distant from the target answers in nearly all cases, the results obtained from WordNet paint a more nuanced picture. Especially when looking at the Wu-Palmer similarity, it becomes clear that a fairly large number of the participants' answers is not completely detached from the target response.

¹ Our description of these measures is based on the excellent explanation in the documentation of Michael Wayne Goodman's Python package *wn*: <https://wn.readthedocs.io/en/latest/api/wn.similarity.html> (last checked 16/11/2024)

² Note that nouns behave differently from verbs and adjectives, as all noun synsets descend from the synset *entity* (see Fellbaum 2006: 666).

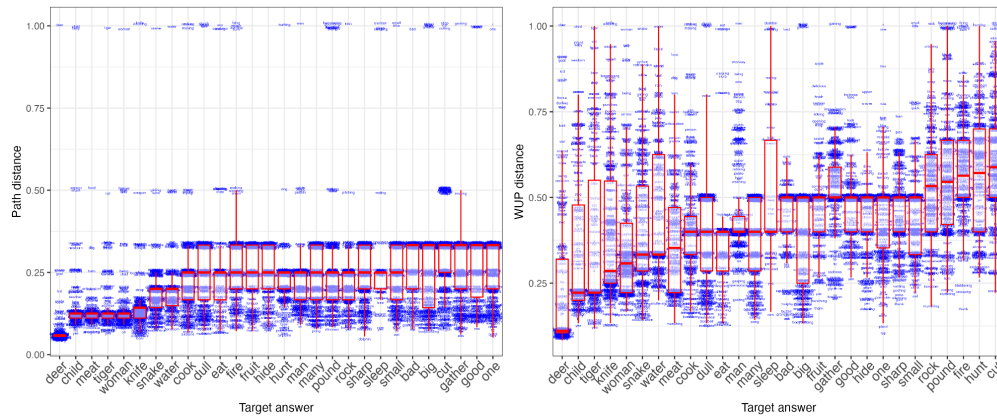


Fig. 1: Path distance (left) and Wu-Palmer distance (right) of participants' answers from the target concepts according to the WordNet taxonomy.

References

Fellbaum, Christiane. 2006. WordNet(s). In *Encyclopedia of Language & Linguistics*, 665–670. Elsevier. <https://doi.org/10.1016/B0-08-044854-2/00946-9>.